

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and maintained **single line diagrams (SLDs)** and **wiring schematics** for low-voltage switchgear panels, ensuring adherence to project specifications and internal design standards.
- Selected and specified electrical components including **circuit breakers**, **busbars**, and **protection relays** for diverse power distribution systems, balancing performance and cost requirements.
- Performed essential electrical calculations such as **load calculations**, **short-circuit analysis**, and **busbar sizing** to guarantee system safety and operational integrity under various conditions.
- Collaborated with **mechanical design teams** to integrate electrical layouts within panel arrangements, optimizing space utilization and manufacturing feasibility.
- Prepared comprehensive technical documentation, including detailed **Bill of Materials (BOM)** and **test procedures**, for manufacturing and quality assurance.
- Supported **factory acceptance tests (FAT)** and initial **site commissioning** phases, troubleshooting design-related issues and ensuring smooth system integration.
- Containerized electrical design applications using **Docker** for consistent development and deployment environments across the engineering team.

SKILLS

Electrical Design – LV Switchgear, Single Line Diagrams (SLDs), Wiring Diagrams, Schematics, Layout Drawings, Panel Arrangements, Bill of Materials (BOM), Test Procedures, Operation & Maintenance Manuals

Electrical Components – Circuit Breakers, Busbars, Protection Relays, Contactors, Control Devices

Electrical Calculations – Load Calculations, Short-Circuit Analysis, Thermal Analysis, Busbar Sizing

Standards & Compliance – IEC 60439, IEC 61439, NSAI IS10101:2025

Design Tools – AutoCAD Electrical, EPLAN

Software & Technologies – Docker, Linux, Shell Scripting, Git

PROJECTS

LV Switchgear Panel Design for Industrial Power Distribution

- Engineered a complete set of electrical schematics and SLDs for a custom LV switchgear panel designed for a large industrial facility, specifying **Siemens circuit breakers** and **ABB busbars**.
- Conducted detailed **short-circuit analysis** and thermal calculations to ensure the selected components and busbar ratings met the stringent IEC 61439 standards for the target load.
- Generated a detailed **Bill of Materials (BOM)** and supported the manufacturing team by clarifying design specifications during panel assembly and wiring.

Automation of Electrical Design Documentation

- Developed Python scripts to automate the generation of repetitive sections of electrical design documentation, such as standard component lists and basic **test procedures**, reducing manual effort by an estimated 15%.
- Integrated script output with **AutoCAD Electrical** drawing templates to streamline the creation of panel layout drawings and wiring diagrams for standard switchgear configurations.
- Applied **Linux shell scripting** to manage file organization and version control for design documentation, improving team collaboration and accessibility.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)